

Complex analysis

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1 Differentiation Theory

2 Integration Theory

2.1 Cauchy integration theorem

3 Functions with singularities

3.1 Taylor and Laurent series

3.2 Poles

Proposition 3.1: Laurent series coefficients calculation

Let f be holomorphic on $B_r(c) \setminus \{c\}$ (for some r) with a pole of order m at a . Then, the Laurent series coefficients of f near c , a_j ($j = -m, -m+1, \dots, -1$), are

$$a_j = \frac{1}{(m+j)!} \left\{ \left(\frac{\partial}{\partial z} \right)^{m+j} (z-c)^m f(z) \right\} \Big|_{z=c}$$

3.3 Residue theorem

Definition 3.1: Residue

Is there anything about essential singularities?

Theorem 3.1: Residue theorem

Let $\{P_1, P_2, \dots, P_n\}$ be poles of a function f defined on U and holomorphic on $U \setminus \{P_1, \dots, P_n\}$. Let $\gamma : [0,1] \rightarrow U$ be a closed curve enclosing all $P_1, \dots, P_n$. Then,

$$\oint_{\gamma} f = 2\pi i \sum_{i=1}^n \text{Res}_f(P_i) \cdot \text{Ind}_{\gamma}(P_i)$$

Remark The condition is for finitely many poles is not because this theorem only works for finitely many poles and fails for infinitely many poles. It also works for the infinitely many poles cases. The condition of finite poles vastly streamlines the theorem.

The intuition of this theorem is: the integration of f on γ that encloses poles $P_1, \dots, P_n$ is equal to $2\pi i$ times of the integration on each small circle around P_i times the winding number of each circle.

When there are infinitely many poles, these poles must accumulate somewhere [details](#).

Proof: Let s_i be the principal parts of Laurent series expansion of f at P_i (near P_i). More explicitly, if the Laurent series expansion of f at P_i is $\sum_{j=-n_i}^{\infty} a_{-j}^i (z - P_i)^j$, then $s_i = \sum_{j=1}^{n_i} a_{-j}^i (z - P_i)^j$.

Consider the function

$$f - \sum_i s_i$$

$f - \sum_i s_i$ is holomorphic on U . [why?](#)

Hence, the integration of it on γ should vanish,

$$\oint_{\gamma} f - \sum_i s_i = 0 \Rightarrow \oint_{\gamma} f = \sum_i \oint_{\gamma} s_i$$

For each i ,

$$\oint_{\gamma} s_i = \oint_{\gamma} \sum_j a_{-j}^i (z - P_i)^{-j} = \sum_{j=1}^{n_i} a_{-j}^i \oint_{\gamma} (z - P_i)^{-j}$$

¹This process is removing all possible singularities of f on U .

From ??, for all $j > 1$, $\oint_{\gamma} (z - P_i)^{-j} = 0$ and for $j = 1$, $\oint_{\gamma} (z - P_i)^{-1} = 2\pi i$. Notice that $\forall i, a_{-1}^i =: \text{Res}_f(P_i)$. Hence,

$$\oint_{\gamma} f = \sum_i a_{-j}^i \cdot 2\pi i = 2\pi i \cdot \sum_i \text{Res}_f(P_i)$$

□

To use residue theorem, we need a quick computation method for residue, as suggested in proposition 3.1, is

$$\text{Res}_f(c) = \frac{1}{(m-1)!} \left\{ \left(\frac{\partial}{\partial z} \right)^{m-1} (z-c)^m f(z) \right\} \Big|_{z=c}$$

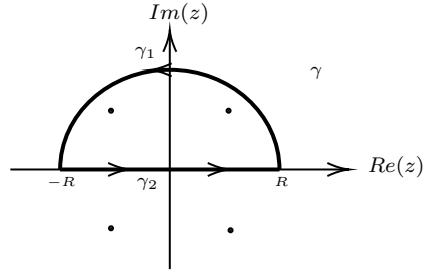
3.3.1 Applications of residue theorem: Calculating integrals

The Residue theorem can be used to calculate some integrals in $\mathbb{R}$:

The question is to evaluate an integral $\int_I f$ with $I \subseteq \mathbb{R}$ and f a real-valued function. The basic idea is find a closed path $\gamma : I \rightarrow \mathbb{C}$ that some part of this path lies on the x -axis, which is related to I . At the same time, the function of real variable should be transferred to a complex-variable form such that its 'projection' onto the real axis is the

$$(1) \int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx = \frac{\pi}{\sqrt{2}}$$

Choose $\frac{z^2}{1+z^4}$. This $\frac{z^2}{1+z^4}$ has four poles, $e^{\frac{\pi}{4}i + \frac{\pi}{2}(n-1)i}$, $n = 1, 2, 3, 4$. Choose the path to be $\gamma := \gamma_1 + \gamma_2$, where $\gamma_1 = Re^{it}$, with $R > 1$ (to enclose the singularities) $t \in [0, \pi]$ and γ_2 is the line on the real line axis from $-R$ to R . Namely, γ is the boundary of upper half disk of radius R centered at zero.



So, this path encloses the poles $e^{\frac{\pi}{4}i}$ and $e^{\frac{3\pi}{4}i}$.

For the integral, one can first do the analysis of itself:

$$\oint_{\gamma} \frac{z^2}{1+z^4} dz = \int_{\gamma_1} \frac{z^2}{1+z^4} dz + \int_{\gamma_2} \frac{z^2}{1+z^4} dz$$

- the second term is $\int_{-R}^R \frac{x^2}{1+x^4} dx$.
- the first term is: $\int_{\gamma_1} \frac{z^2}{1+z^4} dz = \int_0^{\pi} \frac{R^2 e^{2it}}{1+R^4 e^{4it}} Rie^{it} dt$. Then, we do the estimation of the first curve:

$$\begin{aligned} \left| \int_0^{\pi} \frac{R^3 i e^{3it}}{1+R^4 e^{4it}} dt \right| &\leq \int_0^{\pi} \left| \frac{R i e^{3it}}{1+R^4 e^{4it}} \right| dt \\ &\leq \int_0^{\pi} \frac{R^3}{R^4 - 1} dt & |1+R^4 e^{4it}| \geq ||R^4 e^{4it}| - 1| \\ &= \frac{\pi R^3}{R^4 - 1} \end{aligned}$$

Take the limit of the radius of γ to ∞ , i.e. $R \rightarrow \infty$. why? This does not change the value of $\oint_{\gamma} \frac{z^2}{1+z^4} dz$, i.e.

$$\lim_{R \rightarrow \infty} \oint_{\gamma} \frac{z^2}{1+z^4} dz = \oint_{\gamma} \frac{z^2}{1+z^4} dz$$

For the first term, $0 \leq \left| \int_{\gamma_1} \frac{z^2}{1+z^4} dz \right| \leq \frac{\pi R^3}{R^4 - 1}$ and $\lim_{R \rightarrow \infty} \frac{\pi R^3}{R^4 - 1} = 0$. So, $\lim_{R \rightarrow \infty} \left| \int_{\gamma_1} \frac{z^2}{1+z^4} dz \right| = 0$ implies that $\lim_{R \rightarrow \infty} \oint_{\gamma} \frac{z^2}{1+z^4} dz = 0$.

For the second term, $\lim_{R \rightarrow \infty} \int_{\gamma_2} \frac{x^2}{1+x^4} dx =: \int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx$.
Hence,

$$\oint_{\gamma} \frac{z^2}{1+z^4} dz = \int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx$$

From another perspective, the residue theorem 3.1 suggests **winding number**

$$\oint_{\gamma} \frac{z^2}{1+z^4} dz = 2\pi i \cdot \left(\text{Res}_{\frac{z^2}{1+z^4}}(e^{\frac{\pi}{4}i}) + \text{Res}_{\frac{z^2}{1+z^4}}(e^{\frac{3\pi}{4}i}) \right)$$

Let $b_i = e^{\frac{\pi}{4}i + \frac{\pi}{2}(n-1)i}$. Then, $\text{Res}_{\frac{z^2}{1+z^4}}(b_1) = a_1^2(a_1 - a_2)(a_1 - a_3)(a_1 - a_4) = \frac{1}{4}e^{-\frac{\pi}{4}i}$. Similarly, $\text{Res}_{\frac{z^2}{1+z^4}}(a_2) = \frac{1}{4}e^{\frac{5\pi}{4}i}$.
So,

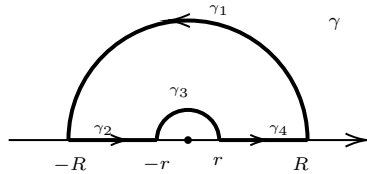
$$\oint_{\gamma} \frac{z^2}{1+z^4} dz = 2\pi i \cdot -\frac{\sqrt{2}}{4}i = \frac{\pi}{\sqrt{2}}$$

(2) **Integration of trigonometric functions:**

$$(2.1) \int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$$

For this, we first choose the complexification of $\frac{\sin x}{x}$ to be $\frac{e^{iz} - e^{-iz}}{z}$ from $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$. **detect the poles** This function has a pole $z = 0$ of order 1. The contour should have some part on the x -axis. As before, we can choose the path γ to be boundary of the upper half disk centered at 0. But, this choice leads to a problem: To integrate $\oint_{\gamma} \frac{e^{iz} - e^{-iz}}{z} dz$, it splits into two parts: one part on γ_1 and another part on γ_2 , where γ_1 and γ_2 are as (1). The integrand is holomorphic on the area enclosed by γ , so $\oint_{\gamma} \frac{e^{iz} - e^{-iz}}{z} dz = 0$. To get the desired integral, we need to push R to ∞ . But, this leads to the integral of $\int_{\gamma_1} \frac{e^{iz} - e^{-iz}}{z} dz$ to ∞ , which because e^{-iz} blows up on the upper half plane when the imaginary part of z goes to infinity.

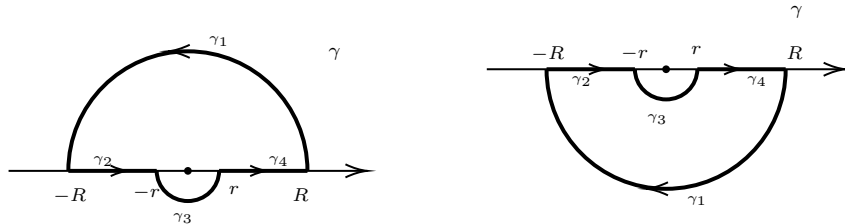
Instead, e^{iz} does not make that happen. At the same time, $\frac{e^{iz}}{z}$ leads to a singularity (pole of order 1) at 0. But, this is the best case. Because the convergence is controlled and the poles can be dealt with by Residue theorem. So, for these reasons, $\frac{e^{iz}}{z}$ is chosen to be the complexification of $\frac{\sin x}{x}$. Correspondingly, the path can avoid or enclose the pole at 0. It is seemingly easier to avoid 0. So, the path γ is chosen to be



$$0 = \oint_{\gamma} \frac{e^{iz}}{z} dz = \int_{\gamma_1} \frac{e^{iz}}{z} dz + \int_{\gamma_2} \frac{e^{iz}}{z} dz + \int_{\gamma_3} \frac{e^{iz}}{z} dz + \int_{\gamma_4} \frac{e^{iz}}{z} dz$$

To estimate $\int_{\gamma_1} \frac{e^{iz}}{z} dz$,

Remark One can also choose different contours such as follows:



$$(2.2) \text{ For } a > 1, \int_0^{\pi} \frac{1}{a + \cos \theta} d\theta = \frac{\pi}{\sqrt{a^2 - 1}}$$

Notice that $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$. $e^{i\theta}$ with $\theta \in [0, \pi]$ is the curve being integrated. So, this is an inverse process of parametrizing an integration. Let $z = e^{i\theta}$. To get a closed curve, another half should be filled to get γe^{it} , where $t \in [0, 2\pi]$. **curve?**

$$\int_0^{\pi} \frac{1}{a + \cos \theta} d\theta = \frac{1}{2} \int_0^{2\pi} \frac{1}{a + \cos \theta} d\theta$$

For the integrand,

$$\frac{1}{a + \cos \theta} = \frac{1}{a + \frac{1}{2} \cdot z + \frac{1}{2z}} = \frac{2z}{z^2 + 2az + 1}$$

and $dz = ie^{i\theta} d\theta = iz d\theta$. Hence,

$$\int_0^\pi \frac{1}{a + \cos \theta} d\theta = -i \oint_\gamma \frac{1}{z^2 + 2az + 1} dz$$

The function has two poles, $\alpha := -a + \sqrt{a^2 - 1}$ and $\beta := -a - \sqrt{a^2 - 1}$, each of order 1. Since $|\alpha| < 1$ and $|\beta| > 1$. Only α is enclosed by the curve γ . The residue of $\frac{1}{z^2 + 2az + 1}$ at α is

$$\text{Res}_{\frac{1}{z^2 + 2az + 1}}(\alpha) = \frac{1}{\alpha - \beta} = \frac{1}{2\sqrt{a^2 - 1}}$$

and by residue theorem,

$$\int_0^\pi \frac{1}{a + \cos \theta} d\theta = -i \oint_\gamma \frac{1}{z^2 + 2az + 1} dz = -i \cdot 2\pi i \cdot \frac{1}{2\sqrt{a^2 - 1}} \cdot 1 = \frac{\pi}{\sqrt{a^2 - 1}}$$

(3) **Integrands with branch points:**

$$(3.1) \int_0^\infty \frac{\log x}{1 + x^2} dx = 0$$

$$(3.2) \text{ For } 0 < c < 1, \int_0^\infty \frac{x^{-c}}{1 + x} dx = \frac{\pi}{\sin \pi c}$$

3.4 Counting zeros and poles

The next is a direct consequence of residue theorem.

Theorem 3.2: The argument principals

A function might be complicated in its form. To know zeros of such a function, it is redundant to look at the whole body of it. Instead, the dominated terms determines all zeros and some other terms are just a noise. The next theorem gives us a method to know zeros of a function without going into details.

Theorem 3.3: Rouché's theorem

Let f and g be two holomorphic functions. If $|f(z)| > |g(z)|$ for all z on the boundary of some closed path C . Then, f and $f + g$ have the same amount of zeros inside the area that enclosed by C .

3.4.1 Application of residue theorems: Summation of series

3.4.2 Application of residue theorems: Jordan's lemma

4 Geometric Theory

4.1 Conformal mapping

4.1.1 Some examples

4.1.2 Riemann mapping theorem

4.1.3 Some comments and applications

4.2 Multivalued functions

4.2.1 Monodromy theorems

4.3 Riemann surfaces

5 Analytic structures and estimates

5.1 Harmonic analysis

Cauchy-Pompiou

5.2 Rigidity theorems

Theorem 5.1: Maximal modulus problem

Theorem 5.2: Hadamard's three line lemma

Theorem 5.3: Montel's theorem

5.3 Some applications to number theory

5.3.1 Riemann-zeta function

6 More topics

6.1 Analytic continuation

6.2 Elliptic functions

7 Riemann surfaces

Definition 7.1: Riemann surfaces

Example (1) Riemann sphere
(2) Complex Torus

7.0.1 Uniformization theorem

The automorphism of Riemann surfaces

Riemann surfaces X	The automorphism group $\text{Aut}(X)$
$\mathbb{S}^2$	$\text{Aut}(\mathbb{S}^2) \cong \left\{ z \mapsto \frac{az+b}{cz+d} : \right\}$
$\mathbb{C}$	$\text{Aut}(\mathbb{C}) \cong \{ z \mapsto az + b : \}$
$\mathbb{D}$	$\text{Aut}(\mathbb{D}) \cong \left\{ z \mapsto e^{i\theta} \frac{az+b}{cz+d} : \right\}$

Figure 1: Automorphism groups for different Riemann surfaces

7.1 Divisors

Let $M(X)$ be a space of meromorphic functions defined on a Riemann surface X . Then, this $M(X)$ is a field because the sum and product of two meromorphic functions is still a meromorphic function.

Divisor is a tool whose intuition is to entail all zeros and poles of a meromorphic function. Define a divisor on a Riemann surface X , denoted $\text{Div}(X)$, is the free abelian group generated by the symbol $[P]$, where P ranges over all zeros and divisors of all functions in $M(X)$.

Let's see the different behaviors of $\text{Div}(X)$ under different Riemann surfaces, say, $X = \mathbb{C} \cup \{\infty\}$ and $X = C/\Lambda$.

For example, if f defined on $\mathbb{C}$ has zeros P, Q and a pole R , each of the multiplicity 2, 1 and 1, respectively. Then, $2[P] + [Q] - [R]$ is a divisor in $\text{Div}(X)$. **on $\mathbb{C}$? on X ?** Correspondingly, the function f has the form

$$f(z) = \frac{(z-P)^2(z-Q)}{z-R}$$

This is the case for $\mathbb{C}$. If we can extend the domain from $\mathbb{C}$ to $\mathbb{C} \cup \{\infty\}$ by one point compactification, then ∞ is also a pole of order 2. The theory lies behind this is changing of variable by $z \mapsto \frac{1}{z}$:

$$\frac{(z-P)^2(z-Q)}{z-R} \mapsto \frac{(\frac{1}{z}-P)^2(\frac{1}{z}-Q)}{\frac{1}{z}-R} = \frac{(1-zP)^2(1-zQ)}{(1-zR)z^2}$$

So, the point at infinity ∞ gets sent to 0.

This extension (extending the domain from $\mathbb{C}$ to $\mathbb{C} \cup \{\infty\}$) balances the number of zeros and the number of poles considering the multiplicity.

Define the order of a function D at some zero or pole P , denoted $\text{ord}_P(D)$. This is a discrete valuation because it satisfies the axioms:

- $\text{ord}_P((fg)) = \text{ord}_P((f)) + \text{ord}_P((g))$.
- $\text{ord}_P((f/g)) = \text{ord}_P((f)) - \text{ord}_P((g))$.
- $\text{ord}_P((f+g)) \geq \min\{\text{ord}_P((f)), \text{ord}_P((g))\}$.

Thus, there is a surjective map $\text{ord}_P : \text{Div}(X) \rightarrow \mathbb{Z}$. It has a kernel whose elements are of order 0 at P , denoted $\text{Div}^0(X) := \ker \text{ord}_P$. So, then we have a short exact sequence

$$0 \rightarrow \text{Div}^0(X) \longrightarrow \text{Div}(X) \xrightarrow{\text{ord}_P} \mathbb{Z} \rightarrow 0$$

We can put an ordering $\geq$ on the divisors $\text{Div}(X)$. **what?** Further, the notion of maximum and minimum elements can be applied. Hence, $\text{Div}(X)$ is a lattice. Now, $\text{Div}(X)$ has two structures and they

Principal divisor: A principal divisor is generated by a single function? Namely, a principal divisor is of the form

$$(f) := \sum_P \text{ord}_P(f) [P]$$

where P ranges all zeros and poles of the function f .

Also, we study the space of functions $f \in M(X)$ that are controlled by a divisor D :

$$L(D) := \{f \in M(X) : (f) + D \geq 0\}$$

This space has the functions whose poles and zeros are dominated by the divisor D . To see this more explicitly, take $D = 2[P] - [Q]$ for example. $(f) + D \geq 0$ means that f has at most a double pole at P and at least a single zero at Q .

In addition, $L(D)$ is a vector space. This means that,

With the vector space in hand, one can decorate each point(divisor) of the lattice with a vector space structure. Hence, this 'decoration' gives a filtration to $\text{Div}(X)$.

There are also some parallel(or monotone) properties for the vector space. Let $D_1, D_2 \in \text{Div}(X)$ be two divisors.

If $D_1 \leq D_2$, then $L(D_1) \leq L(D_2)$ as a subspace.

If $D_1 \sim D_2$, then $L(D_1) \cong L(D_2)$ and further $\dim L(D_1) = \dim L(D_2)$.

If $D_1 < 0$, then $L(D_1) \cong \{0\}$.

By adding a divisor, (P) , **P?**, the dimension of the new space $L(D + (P))$ cannot exceed $\dim L(D)$ too many. In fact, it at most reach a lifting of the dimension by 1. In other words, the codimension of

For $X = \mathbb{C}/\Lambda$, the case differs. there is a map

$$0 \rightarrow \text{PDiv}(X) \longrightarrow \text{Div}(X) \xrightarrow{(\text{ord}_P, \text{sum}_P)} \mathbb{Z} \times \mathbb{C}/\Lambda \rightarrow 0$$

Since two numbers related to a divisor D , $\deg_P(D)$ and $\dim L(D)$ are at hand, it is quite natural to ask that how they could be related. To feel it, let's try to calculate two numbers for two different Riemann surfaces.

For $X = \hat{\mathbb{C}}$,

$\deg_P(D)$	$\dim L(D)$
-3	
-2	
-1	
0	
1	
2	
3	
...	

Figure 2: $\deg_P(D)$ and $\dim L(D)$ for $X = \hat{\mathbb{C}}$

For $X = \mathbb{C}/\Lambda$,